

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE PHYSICS

Paper 2

Friday 30 May 2025

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- a protractor
- the Physics Equations Sheet (enclosed).

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information

- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



J U N 2 5 9 2 0 3 2 0 1

Answer **all** questions in the spaces provided.

0 1

Thinking distance and braking distance both affect the stopping distance of a car.

0 1 . 1

Which **two** factors affect the thinking distance of a car?

[2 marks]

Tick (✓) **two** boxes.

The condition of the car's brakes

☐

The condition of the car's tyres

☐

The driver being distracted

☐

The steepness of the road

☐

The tiredness of the driver

☐

0 1 . 2

What factor affects **both** the thinking distance **and** the braking distance?

[1 mark]

Tick (✓) **one** box.

The driver taking drugs

☐

The mass of the car

☐

The speed of the car

☐

The weather conditions

☐


0 1 . 3 Ice on the road surface increases the braking distance.

Which force between the tyres and the road surface would be reduced by the ice?

[1 mark]

Tick (✓) **one** box.

Air resistance

☐

Friction

☐

Gravitational force

☐

Normal contact force

☐

0 1 . 4 The brakes of a car are applied so the car stops safely.

The deceleration of the car is 3.0 m/s^2 .

The mass of the car is 1200 kg.

Calculate the resultant force acting on the car during braking.

Use the equation:

$$\text{resultant force} = \text{mass} \times \text{deceleration}$$

[2 marks]

Resultant force = _____ N

Question 1 continues on the next page

Turn over ►



0 1 . 5

Why is there a safety risk if the deceleration of the car is too large?

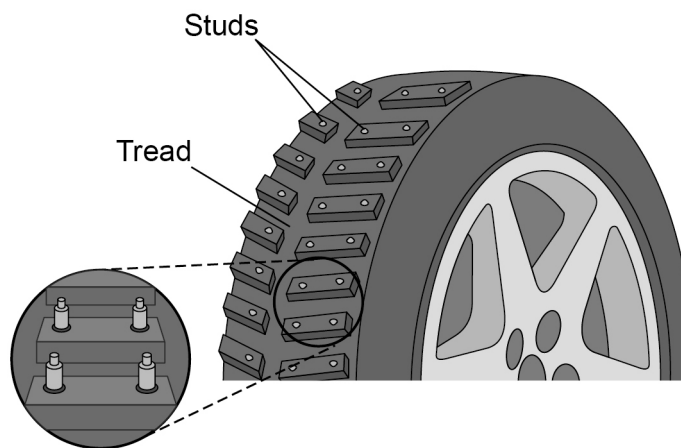
[1 mark]

0 1 . 6

Snow tyres can be used on cars travelling on roads covered in snow.

Figure 1 shows a snow tyre.

Figure 1



The snow tyre has small pieces of metal called studs that stick out from the surface of the tyre.

Why do snow tyres reduce the braking distance of the car when travelling on roads covered in snow?

[1 mark]



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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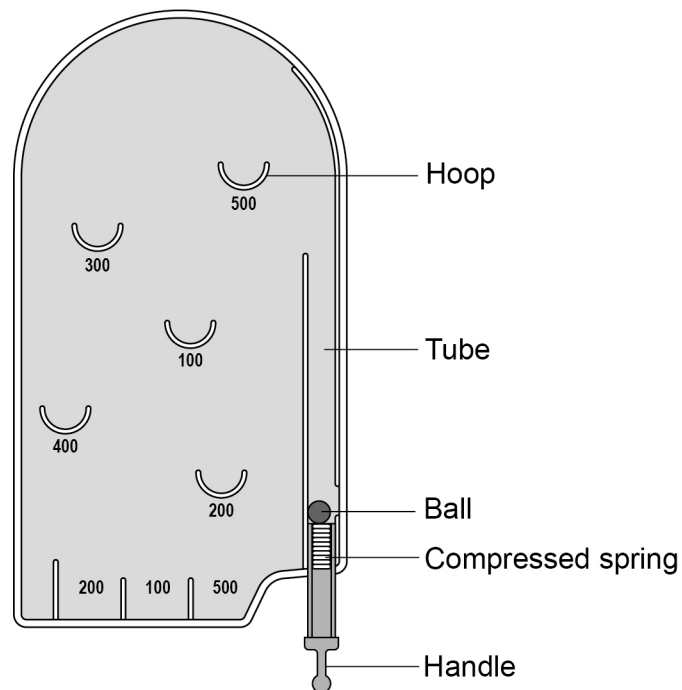


0 5

0 2

Figure 2 shows a pinball toy used by a child.

Figure 2



The child pulls the handle down to compress the spring.

When the handle is released, the spring returns to its original length and the ball moves vertically upwards.

The ball falls into a hoop to score points.

0 2 . 1

The spring constant of the spring is 200 N/m.

The compression of the spring is 0.025 m.

Calculate the force on the spring.

Use the equation:

$$\text{force} = \text{spring constant} \times \text{compression}$$

[2 marks]

Force = _____ N



0 2 . 2

The child releases the handle and the ball moves vertically upwards.

The change in gravitational potential energy of the ball is 0.0049 J.

The mass of the ball is 0.0050 kg.

Calculate the change in vertical height of the ball.

gravitational field strength = 9.8 N/kg

Use the Physics Equations Sheet.

[3 marks]

Change in vertical height = _____ m

0 2 . 3

The child returns the ball to the position shown in **Figure 2**.

Explain how the vertical height the ball reaches can be changed by the child.

Use **Figure 2** and ideas about energy in your answer.

[4 marks]



0 3

Refraction occurs when a ray of light passes from one material to another material.

0 3 . 1

What causes refraction?

[1 mark]

Tick (✓) **one** box.

A change in the amplitude of the light wave.

☐

A change in the frequency of the light wave.

☐

A change in the speed of the light wave.

☐

A student investigated how the angle of incidence affects the angle of refraction for a ray of light entering a glass block.

Table 1 shows the results.

Table 1

Angle of incidence in degrees	Angle of refraction in degrees
30	18
40	23
50	28
60	32
70	35
80	37



[6 marks]

[illegible]

Turn over ►



0	3	.	3
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When the angle of incidence was 40° , the angle of refraction was 23° .

Calculate the refractive index of the glass.

Use the Physics Equations Sheet.

Give your answer to 2 significant figures.

[3 marks]

Refractive index (2 significant figures) = _____

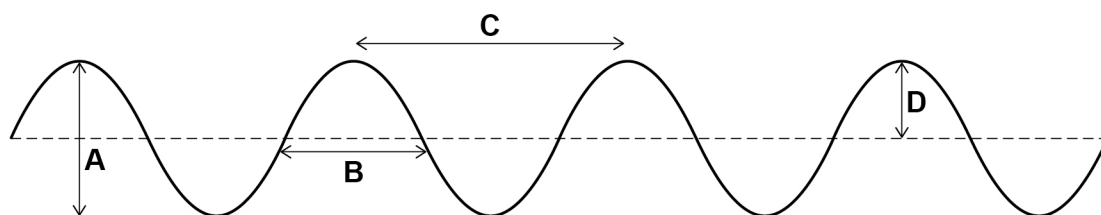
10



0 4

Figure 3 shows a diagram of a wave.

Figure 3



0 4 . 1

Which label in **Figure 3** shows the wavelength of the wave?

[1 mark]

Tick (✓) **one** box.

A

☐

B

☐

C

☐

D

☐

0 4 . 2

Which label in **Figure 3** shows the amplitude of the wave?

[1 mark]

Tick (✓) **one** box.

A

☐

B

☐

C

☐

D

☐

Question 4 continues on the next page

Turn over ►



Infrared radiation is an electromagnetic wave.

The range of frequencies for infrared radiation is 3.0×10^{11} Hz to 4.3×10^{14} Hz.

0 4 . 3

What is meant by 'range'?

[1 mark]

0 4 . 4

What is meant by frequency of a wave?

[1 mark]



A student used an infrared detector to measure the amount of infrared radiation emitted from a container at different temperatures.

This is the method used.

- 1 Use a measuring cylinder to pour 100 ml of hot water into the container.
- 2 Use a thermometer to measure the temperature of the water.
- 3 Place the infrared detector 2.0 cm from the container.
- 4 Measure the amount of infrared radiation emitted from the container in 10 s.
- 5 Repeat steps 1 to 4 using different temperatures of water.

0 4 . 5

The student used a black container.

Why was a black container used rather than a white container?

[1 mark]

Tick (✓) **one** box.

Black surfaces are better emitters of infrared radiation.

☐

Black surfaces are better reflectors of infrared radiation.

☐

White surfaces are better absorbers of infrared radiation.

☐

White surfaces are better transmitters of infrared radiation.

☐

0 4 . 6

What is the relationship between the temperature of the water and the frequency of the infrared radiation emitted?

[1 mark]

Tick (✓) **one** box.

The lower the temperature of water, the greater the frequency of the infrared radiation.

☐

All temperatures of water emit infrared radiation at the same frequency.

☐

The higher the temperature of water, the greater the frequency of the infrared radiation.

☐

Question 4 continues on the next page

Turn over ►



0 4 . 7

The wavelength of the infrared wave is 0.75 mm.

speed of infrared wave = 3.0×10^8 m/s

Calculate the frequency of the infrared wave.

Use the Physics Equations Sheet.

[3 marks]

Frequency = _____ Hz

0 4 . 8

Infrared cameras can be used to locate people when it is dark.

Why are infrared cameras used instead of cameras that detect visible light waves?

[1 mark]

10



0 5

Stars go through a lifecycle.

0 5 . 1

Stars are formed from dust and gas.

What force pulls the dust and gas together?

[1 mark]

0 5 . 2

Name the process in the core of the star that releases energy.

[1 mark]

0 5 . 3

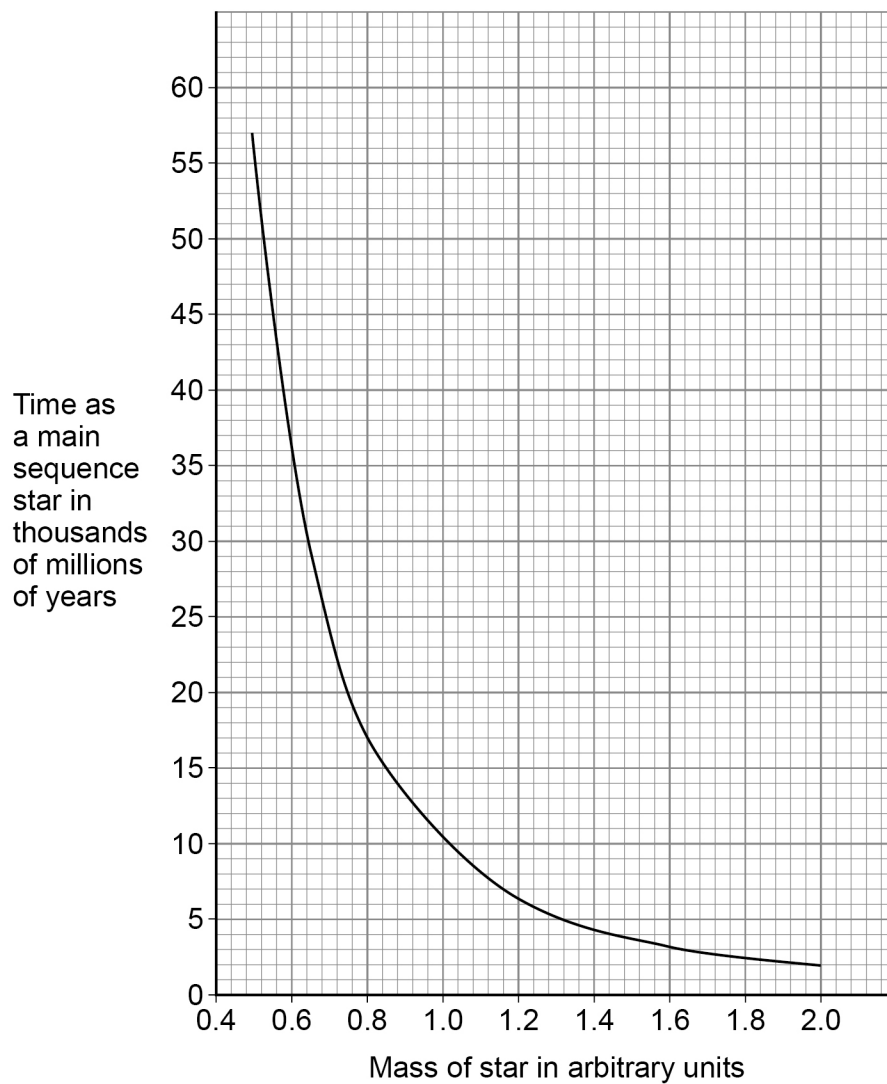
How do the forces in a star compare when the star is in the main sequence?

[1 mark]

Question 5 continues on the next page**Turn over ►**

Figure 4 shows how the time as a main sequence star is affected by the mass of the star.

Figure 4



0 5 . 4

The Sun has a mass of 1.0 arbitrary unit.

How much time will the Sun spend as a main sequence star?

[1 mark]

Time = _____ thousands of millions of years



0 5 . 5

Describe the relationship between the time as a main sequence star and the mass of the star.

[2 marks]

0 5 . 6

Star **X** and star **Y** have different life cycles.

Star **X** will end its life cycle as a black hole.

Star **Y** will end its life cycle as a black dwarf.

Explain why star **X** and star **Y** will be main sequence stars for different amounts of time.

Use **Figure 4**.

[2 marks]

0 5 . 7

Explain why the Sun will **not** form elements heavier than iron during its life cycle.

[2 marks]



0 6

Five swimmers swam the same distance in a swimming race.

Table 2 shows the times recorded for the swimmers.

Table 2

Swimmer	Time in seconds
A	128.34
B	123.46
C	125.00
D	128.29
E	129.55

0 6 . 1

Which swimmer in **Table 2** had the fastest speed?

[1 mark]

Tick (✓) **one** box.

Swimmer **A**

☐

Swimmer **B**

☐

Swimmer **C**

☐

Swimmer **D**

☐

Swimmer **E**

☐

0 6 . 2 Swimmer **C** had an average speed of 1.6 m/s.

Determine the distance of the race.

Use **Table 2** and the Physics Equations Sheet.

[3 marks]

Distance = _____ m

Touch-boards are placed at the end of the swimming pool.

At the beginning of the race a timer is started.

When the swimmer pushes the touch-board, the timer is stopped.

The resolution of the timer is 0.01 s.

0 6 . 3 What is meant by 'resolution'?

[1 mark]

0 6 . 4 Why would a timer with a resolution of 0.1 s **not** be suitable for this race?

Use **Table 2**.

[1 mark]

Question 6 continues on the next page

Turn over ►



0 6 . 5

Two methods are used to time each swimmer in the race.

Both use a starting device which makes a loud sound when pressed.

Method 1

- The starting device is connected to the timer.
- When the starting device is pressed, the timer starts.
- The timer is stopped when the swimmer touches the touch-board at the end of the pool.

Method 2

- A person starts a stopclock when the sound from the starting device is heard.
- The person stops the stopclock when they see the swimmer reach the end of the pool.

The stopclock has the same resolution as the timer.

Explain why **method 1** is more accurate than **method 2** in this race.

[3 marks]



As the swimmer comes close to the touch-board, a small amount of water hits the touch-board.

The kinetic energy of the water that hits the touch-board is 0.090 J.
The mass of water is 80 g.

The water comes to rest on the touch-board in a time of 2.5 s.
The force needed to stop the timer is 18 N.

Determine if the water hits the touch-board with enough force to stop the timer.

[5 marks]

[illegible]

Conclusion _____

14

Turn over for the next question

Turn over ►



0 7

Nuclear power stations contain nuclear reactors.

Graphite is a material that is used to absorb neutrons and control fission reactions in a nuclear reactor.

The graphite contains isotopes of carbon.

Figure 5 shows two isotopes of carbon.

Figure 5



0 7 . 1

Describe **one** similarity and **one** difference between the nucleus of a carbon-12 atom and the nucleus of a carbon-14 atom.

[2 marks]

Similarity _____

Difference _____

0 7 . 2

A carbon-14 nucleus is unstable and emits a beta particle when it decays.

Describe the process in the nucleus that produces a beta particle.

[2 marks]



07.3

Complete the decay equation for carbon-14 when it decays into nitrogen (N).

[2 marks]



07.4

After the graphite is used in the nuclear reactor, it contains a large amount of radioactive carbon-14.

The half-life of carbon-14 is 5700 years.

Explain why the graphite has to be disposed of safely.

[2 marks]

Question 7 continues on the next page

Turn over ►



07.5

Scientists have suggested using carbon-14 to make batteries.

The most common type of battery used in small electrical appliances is an AA battery.

Table 3 shows the characteristics of a carbon-14 battery and an AA battery.

Table 3

	Total energy in joules	Maximum power in watts	Lifetime in years
Carbon-14 battery	33 000 000 000 000	0.0000010	5700
AA battery	10 000	3.0	2

Evaluate the use of the different batteries in **Table 3**.

[4 marks]

Extra space

12



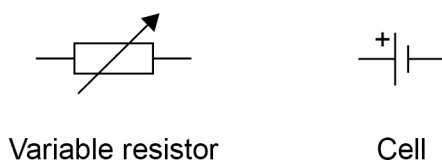
0 8

A student investigated how the potential difference across a thermistor affected the current in the thermistor.

The thermistor was kept at a constant temperature.

Figure 6 shows circuit symbols for some of the equipment used in the investigation.

Figure 6

**0 8****. 1**

The student used the components in **Figure 6** and:

- an ammeter
- a switch
- a thermistor
- a voltmeter.

Draw a circuit diagram for a circuit the student could have used.

[3 marks]

Question 8 continues on the next page

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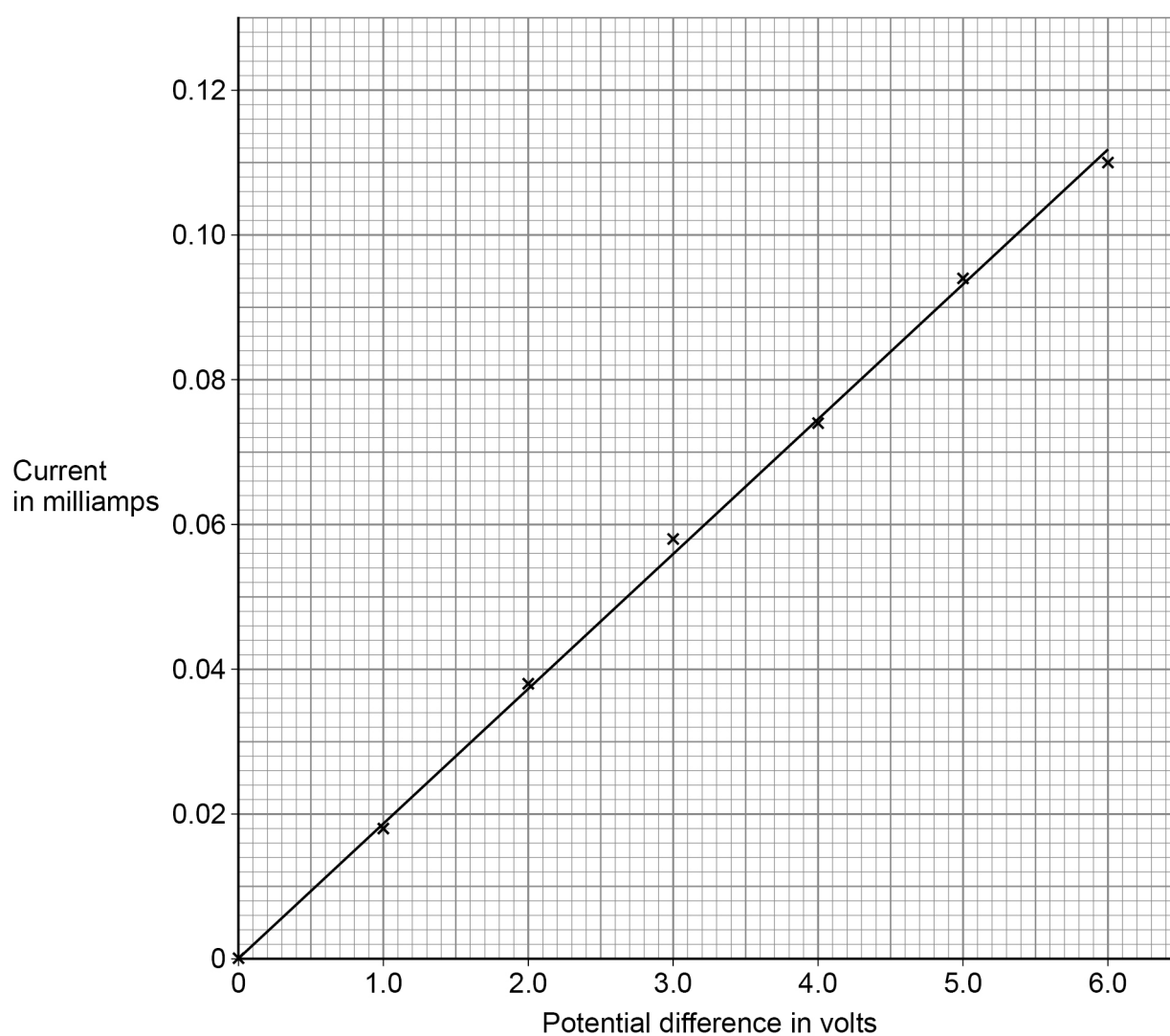
0 8 . 2

Why was a variable resistor needed in the investigation?

[1 mark]

Figure 7 shows the results of the investigation for the thermistor at constant temperature.

Figure 7



0 8 . 3

What was the current in the thermistor when the potential difference across the thermistor was 1.5 V?

Give your answer in milliamps and standard form.

[2 marks]

Current (in standard form) = _____ mA

0 8 . 4

Explain why it is important to use a small current in this investigation.

[2 marks]

0 8 . 5

Describe the relationship shown in **Figure 7**.

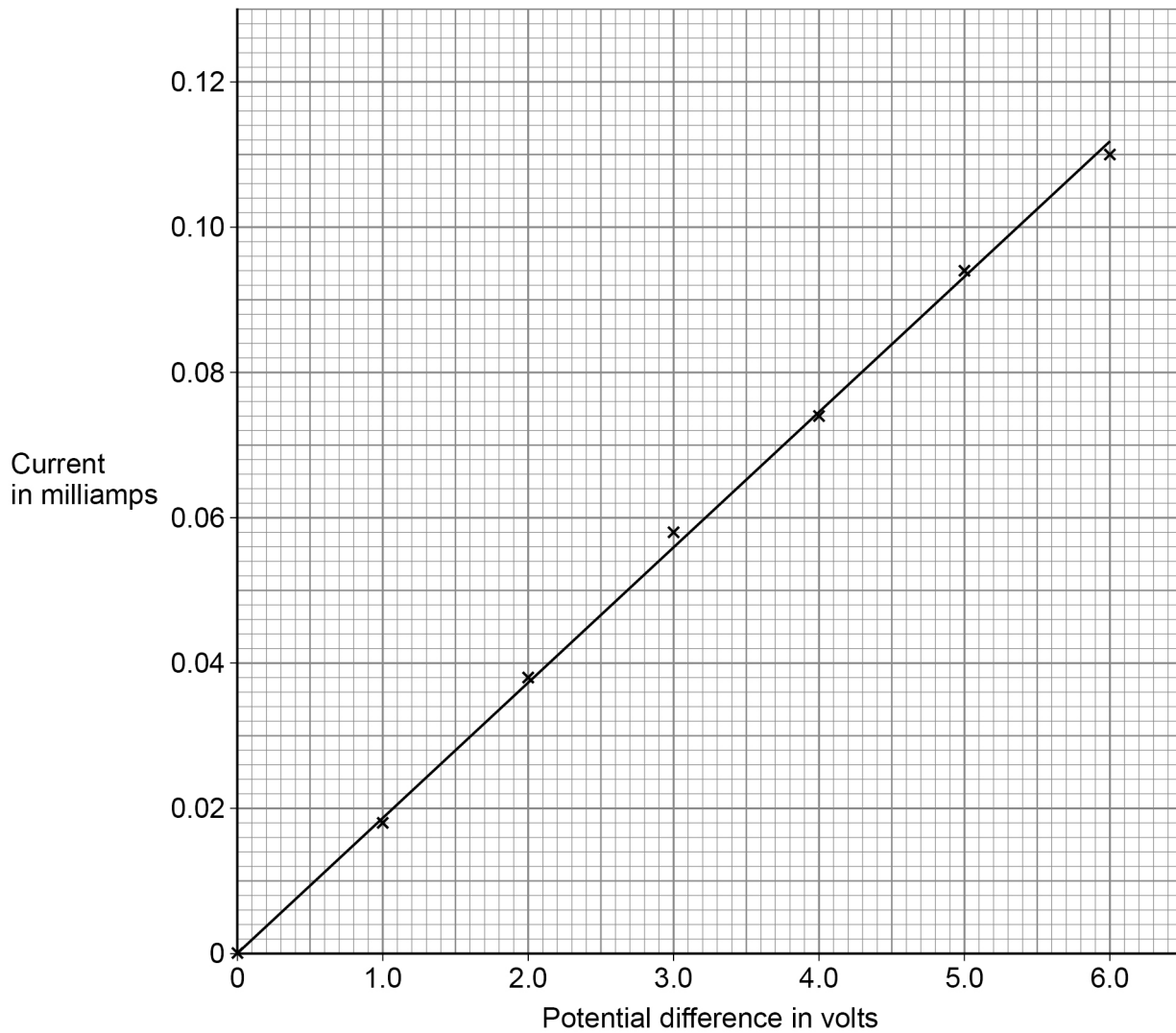
[2 marks]

Question 8 continues on the next page

Turn over ►

Figure 7 is repeated below.

Figure 7



0 8 . 6

A different circuit was set up by the student.

The thermistor was at the same temperature as before.

The thermistor was connected in parallel with a cell and a resistor.

The potential difference across the cell was 4.3 V.

The current in the cell was 0.13 mA.

Determine the resistance of the resistor.

Use **Figure 7**.

[5 marks]

Resistance = _____ Ω

0 8 . 7

Explain how the graph in **Figure 7** would be different if the thermistor was **not** kept at a constant temperature.

[2 marks]

END OF QUESTIONS



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